


# IIT Madras

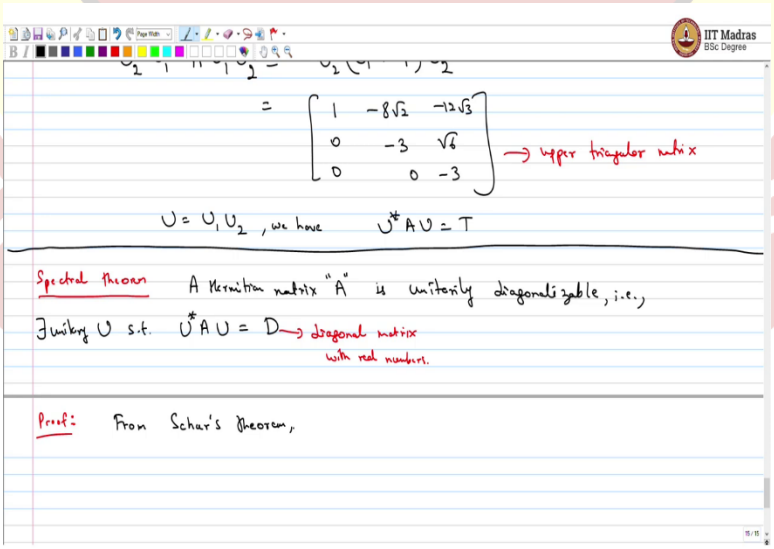
## ONLINE DEGREE

**Machine Learning Foundations**
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**Indian Institute of Technology, Madras**
**Diagonalization of Hermitian Matrices (Part 2)**

Welcome to lecture 5 in week 5 of machine learning foundations. We are continuing the discussion on the linear algebra side of things. In this particular lecture, we will be able to prove the most important theorem that we wanted to get to which is the spectral theorem, which in essence states the following a Hermitian matrix is unitarily diagonalizable. Do note that this is not an if and only if. So, if a matrix is unitarily diagonalizable that does not imply that matrix is necessarily Hermitian. All we are saying is that if a matrix is Hermitian, then it is unitarily diagonalizable.

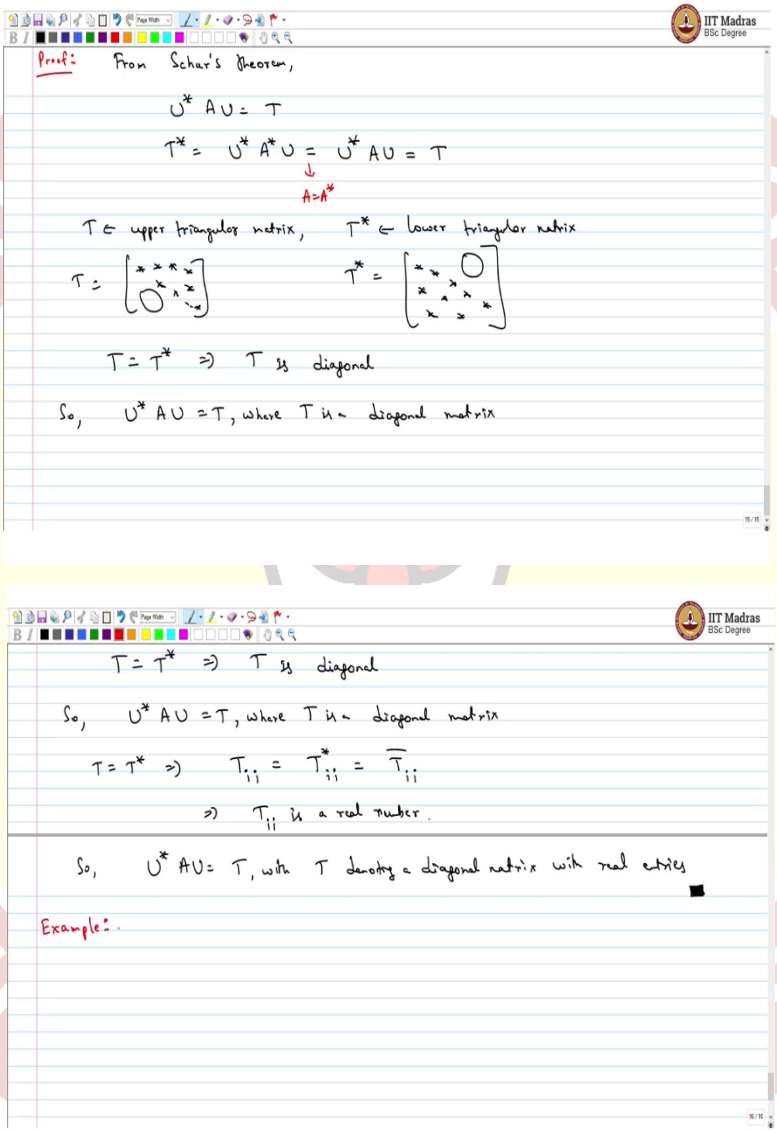
Now, an important corollary of this fact in the context of real vector spaces is that a real symmetric matrix is orthogonally diagonalizable. That is the version of spectral theorem for real symmetric matrices. We will prove the spectral theorem for complex matrices for Hermitian matrices by using what we proved in the previous lecture, which is Schur's theorem which actually stated that every matrix is upper triangularizable. We use this fact to show that a Hermitian matrix is unitarily diagonalizable and then a few examples after that.

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Now, the main theorem so called spectral theorem. The most general case says that any Hermitian matrix is unitarily diagonalizable that is U *, that is there exists unitary matrix U such that, a Hermitian matrix A is unitarily diagonalizable that is U * AU is diagonal matrix with real numbers.

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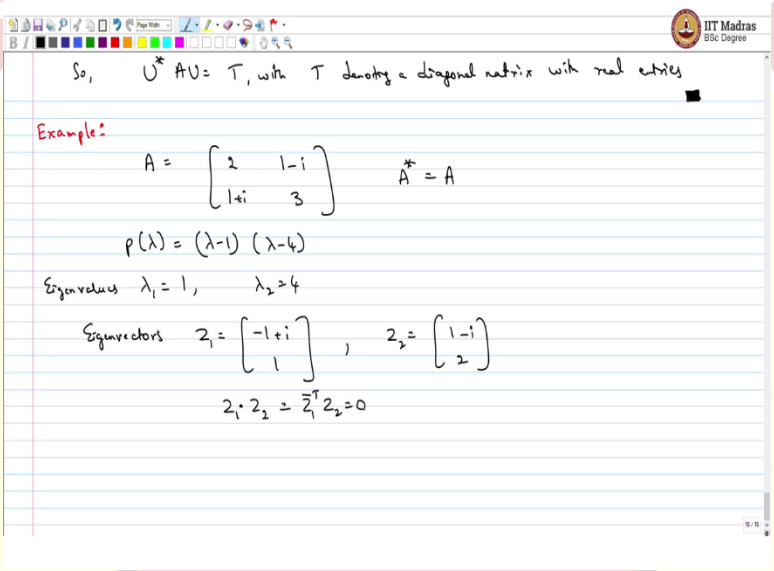


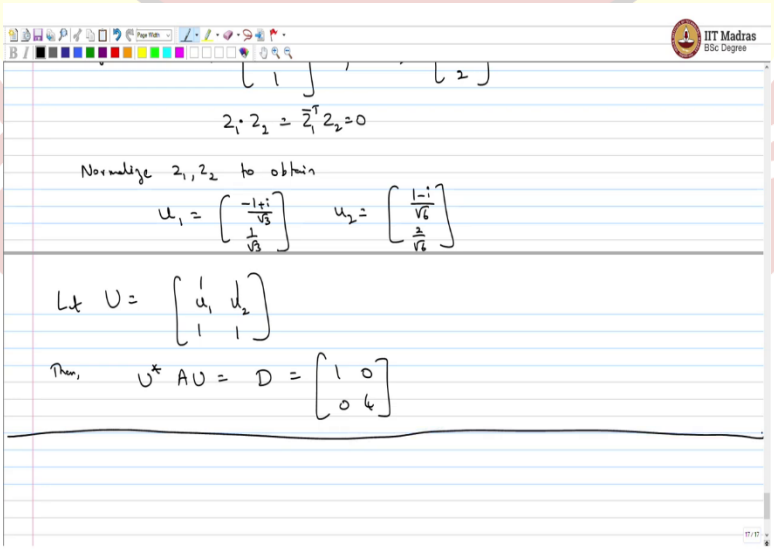

The proof is very simple given Schur's theorem, from Schur's theorem what we have is we have U * AU is a upper triangular matrix. Now, let us look at T *. This is U * A * U, but this is the same as U * AU because A = A *. It is a Hermitian matrix. Now, but U * AU is just T just by construction. Now, T is a upper triangular matrix. So, what is T *? T * would be a lower triangular

matrix. So, if you have T here, T, you have 0 here, then when you take its conjugate transpose, T * which just move all the entries above the diagonal below.

Now, if T = T * that can only happen if T is diagonal. So, we have, T is a diagonal matrix. So, U * AU is equal to T where T is a diagonal matrix. This is not all. We also claim that the diagonal entries are just real. So, let us also show that, now, T = T * implies, if you consider just the diagonal entries, Tii = T * ii. This is the, but this is the same as T bar ii implies Tii is a real number. So, the final claim is U * AU = T with T denoting a diagonal with real entries. So, that is where proof.

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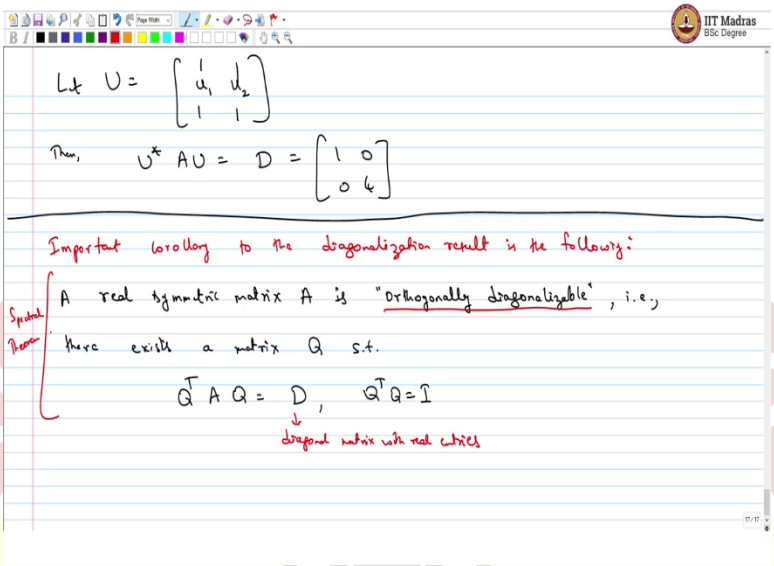


So, let us look at an example. It is easy to check that A * = A. What is the characteristic polynomial of A? P of $\lambda = \lambda - 1$ into $\lambda - 4$ and so the Eigen value $\lambda$ 1 = 1, $\lambda$ 2 = 4. It is a very simple matrix. I am just going to write down the Eigen vectors. Let us call the Z1 = - 1 + i equal to 1 corresponding to $\lambda$ 1 and Z2 = 1 - i. It is easy to check that Z1 transpose Z2 = 0. But we need a unitary matrix, so we need to do a little more work.

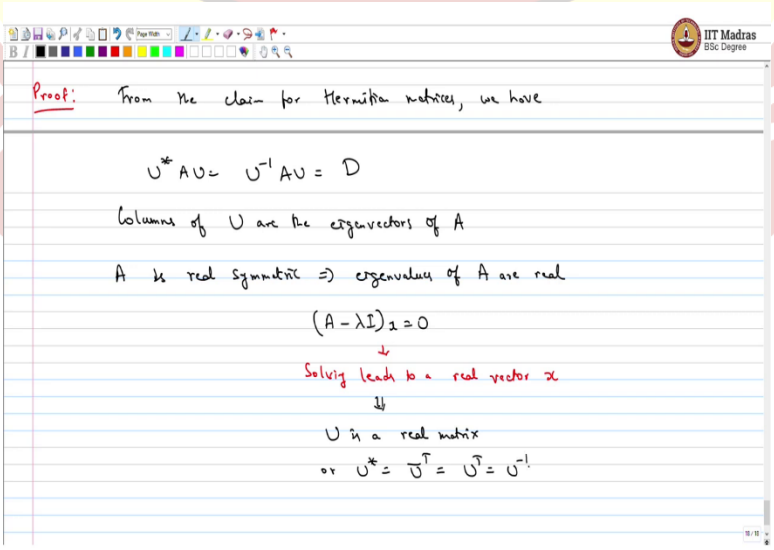
So, just normalize, so normalize Z1 comma Z2 to obtain U1 is just a normalized version, so it is - 1 + i divided by √3. U2 = 1 - i divided by √6 and 2 divided by √6. And then let U = u1, u2. Then, what is U * AU? U * AU can be verified to be the diagonal matrix this matrix.

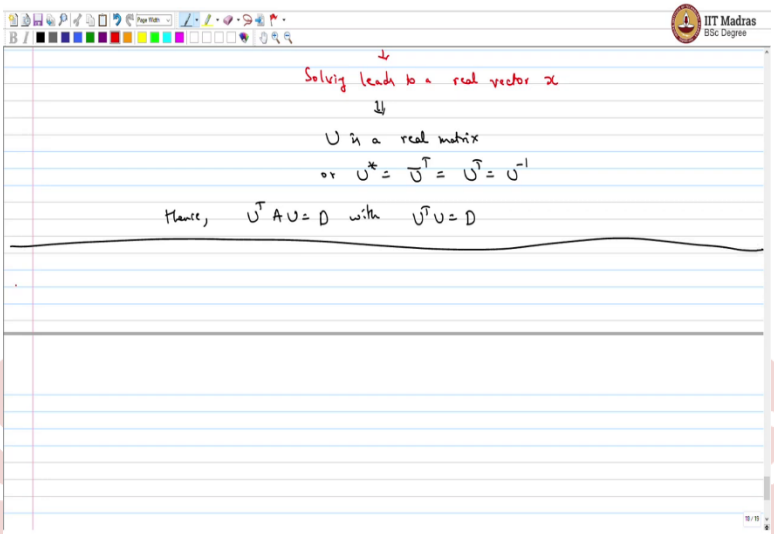
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A very important corollary to the result above, important corollary to the diagonalization result is the following. So, the real world counter parts of Hermitian matrix is a real symmetric matrix A is, and the world, in the real world instead of the unitary concept in a complex vector space is orthogonal, orthogonally diagonalizable that is there exists a matrix Q such that, instead of U * AU I will take Q transpose AQ and I will get a diagonal matrix, Q transpose Q is identity, because that is a orthogonal matrix and this is a diagonal matrix with real entries. This is the spectral theorem.

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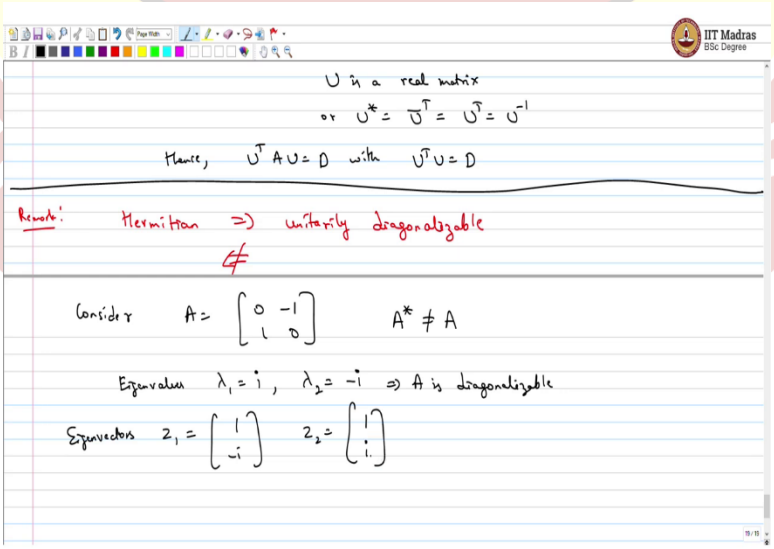


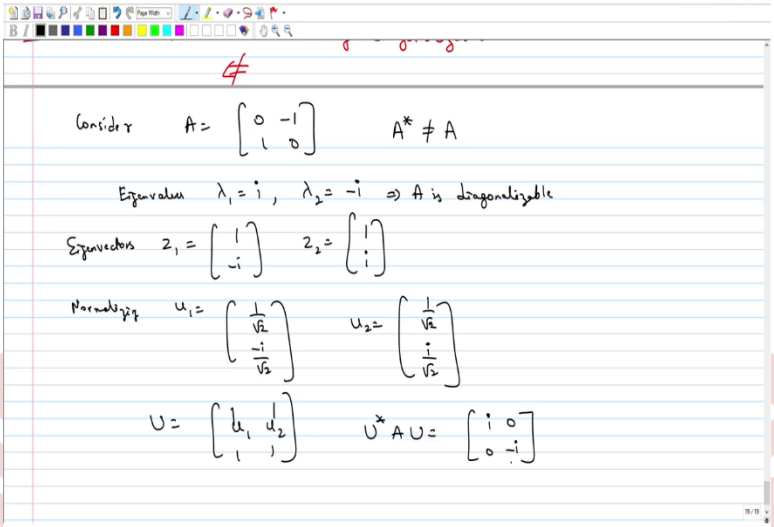



And the proof is fairly straightforward. I will just give it to you. From the claim for Hermitian matrices we have U * AU = U inverse AU = D, a diagonal matrix. And columns of this matrix U are the eigenvectors of A that is given. Now, it is clear. A is real symmetric, which is same that it is Hermitian implies Eigen values of A are real.

So, if I look at A - $\lambda$ I × x = 0, now solving leads to a real vector x. And this implies U is a real matrix or U * = U bar transpose = U transpose = U inverse. That is all that is there. Hence we have U transpose AU = diagonals with U transpose U = D.

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I will end this topic with one final remark which is the following. We have Hermitian matrices are, if a matrix is Hermitian implies it is unitarily diagonalizable, but this does not mean that the only unitarily diagonalizable matrices are Hermitian matrices, particular the converse is not true. So, to see this, consider a simple matrix such as 0, - 1, 1, 0. It is easy to see that A * is simply not equal to A, but you can always find enough eigenvectors.

So, Eigen values under $\lambda$ 1 = i, you can just check this, $\lambda$ 2 = - i this already implies distinct Eigen values implies A is diagonalizable and the corresponding eigenvectors are Z1 is 1 - i, Z2 is 1 i. We can normalize it. Normalizing just two vectors so it is easy, so U1 is just divided by the length of the vector so 1 by √2 - i by √2 and U2 is 1 by √2, i by √2. All you have to do is write this matrix U, u1, u2 and you can check that U * AU is i, 0, 0, - i.

A recap of what happened in today's lecture. What did we learn today, the first thing that we understood today was we could state and prove the following claim which is a Hermitian matrix is unitarily diagonalizable. Now, this result in the context of real vector spaces states that a real symmetric matrix is orthogonally diagnosable, which is a very important result also, which is well known as the spectral theorem.

And now, we did a few examples to verify this fact for a complex matrix as well as a real symmetric matrix. In the complex case, we took a Hermitian matrix and we could unilaterally diagonalize it. And lastly, we actually showed, exhibited a matrix which is unitary diagonalizable, but which is not Hermitian. So, please do understand the claim well. If a matrix Hermitian it is unitarily

diagonalizable, but the converse is not true. So, there are matrices which are not Hermitian but still unitarily diagonalizable.